$\theta > 0$ T1

$\xi \sim R(0, \theta)$

1) $\hat{\theta}_1 = 2\bar{x} = 2\frac{1}{n}\sum_{i=1}^{n} x_i$

$$M\xi = \int_{-\infty}^{\infty} x \, dF(x,\theta) = \int_0^{\theta} x \frac{1}{\theta} dx = \frac{\theta}{2}$$

$$dF(x,\theta) = p(x,\theta) = \frac{1}{\theta}$$

$$M\xi^2 = \int_{-\infty}^{\infty} x^2 dF(x,\theta) = \int_0^{\theta} x^2 \cdot \frac{1}{\theta} dx = \frac{\theta^2}{3}$$

$$D\xi = M\xi^2 - (M\xi)^2 = \frac{\theta^2}{3} - \frac{\theta^2}{4} = \frac{\theta^2}{12}$$

$$M\hat{\theta}_1 = M\frac{2}{n}\sum X_i = \frac{2}{n} \cdot \sum MX_i = \frac{2}{n} \cdot n \cdot M\xi = 2M\xi = \theta$$

$\Rightarrow \hat{\theta}_1$ - несмещенная

$$D\hat{\theta}_1 = D\left(\frac{2}{n}\sum_{i=1}^{n} X_i\right) = \frac{4}{n^2} \cdot D\left(\sum X_i\right) = \frac{4}{n^2} \cdot \sum DX_i = \frac{4}{n^2} \cdot n \cdot D\xi =$$

$$= \frac{4}{n} \cdot D\xi = \frac{4}{n} \cdot \frac{\theta^2}{12} = \frac{\theta^2}{3n} \xrightarrow[n\to\infty]{} 0 \Rightarrow$$ состоятельная - $\hat{\theta}_1$

2) $\hat{\theta}_2 = X_{min}$ $\quad \xi \sim F(x)$

$$\xi_{min} \sim 1 - (1 - F(x))^n \qquad \varphi(x) = n(1 - F(x))^{n-1} \cdot F'(x) =$$

$$= n\left(1 - \frac{x}{\theta}\right)^{n-1} \cdot \frac{1}{\theta} \{(0,\theta)\}$$

$$MX_{min} = \int_{-\infty}^{\infty} X \, dF(x,\theta) = \int_0^{\theta} X \cdot n\underbrace{\left(1 - \frac{x}{\theta}\right)}_{t}^{n-1} \cdot \frac{1}{\theta} dx \overset{\ominus}{=}$$

$x = \theta(1-t)$

$$\ominus = -\int_1^0 \theta n(1-t) t^{n-1} dt = n\theta \int_0^1 (t^{n-1} - t^n) dt = n\theta\left(\frac{1}{n} - \frac{1}{n+1}\right) = \frac{\theta}{n+1}$$

смещенная оценка

$\widetilde{\Theta}_2' = (n+1)\, X_{min}$

$M\widetilde{\Theta}_2' = \frac{\Theta}{n+1} \cdot (n+1) = \Theta$ несмещенная оценка

$M\widetilde{\Theta}_2' = M X_{min} \cdot (n+1)$

$$M X_{min}^2 = \int_0^\Theta x^2\, n \cdot \underbrace{\left(1-\frac{x}{\Theta}\right)}_{t}^{n-1} \cdot \frac{1}{\Theta}\, dx = -\int_1^0 n\Theta^2 (1-t)^2 t^{n-1} \cdot \frac{1}{\Theta} \cdot \Theta\, dt =$$

$x = \Theta(1-t)$

$1 - \frac{x}{\Theta} = t \Rightarrow \frac{x}{\Theta} = 1-t$

$$= \int_0^1 n\Theta^2 (1-t)^2 t^{n-1}\, dt = n\Theta^2 \int_0^1 (1-2t+t^2) t^{n-1}\, dt = n\Theta^2 \int_0^1 (t^{n-1} - 2t^n + t^{n+1})\, dt = n\Theta^2 \cdot \left(\frac{t^n}{n} - 2\frac{t^{n+1}}{n+1} + \frac{t^{n+2}}{n+2}\right)\Big|_0^1 = n\Theta^2\left(\frac{1}{n} - \frac{2}{n+1} + \frac{1}{n+2}\right)$$

$$= n\Theta^2\left(\frac{(n+1)(n+2) - 2n^2 - 4n + n^2 + n}{n(n+1)(n+2)}\right) = n\Theta^2 \cdot \frac{n^2+3n+2-2n^2-4n+n^2+n}{n(n+1)(n+2)} =$$

$$= \frac{2\Theta^2}{(n+1)(n+2)}$$

$$D\widetilde{\Theta}_2' = D((n+1)X_{min}) = (n+1)^2 D X_{min} = (n+1)^2 (M X_{min}^2 - (M X_{min})^2) =$$

$$= (n+1)^2 \left(\frac{2\Theta^2}{(n+1)(n+2)} - \frac{\Theta^2}{(n+1)^2}\right) = (n+1)^2 \cdot \frac{2\Theta^2(n+1) - \Theta^2(n+2)}{(n+1)^2(n+2)} =$$

$$= \frac{n \cdot \Theta^2}{n+2} \underset{n\to\infty}{\not\to} 0 \Rightarrow \text{не работ. достаточ. условие}$$

$$\forall \Theta > 0 \quad \forall \varepsilon > 0 \quad P(|\widetilde{\Theta}_2' - \Theta| \geq \varepsilon) \xrightarrow[n\to\infty]{} 0$$

$$P(|\widetilde{\Theta}_2' - \Theta| \geq \varepsilon) = P(\widetilde{\Theta}_2' \geq \Theta + \varepsilon) + P(\widetilde{\Theta}_2' \leq \Theta - \varepsilon)$$

$$P(|\widetilde{\Theta}_2' - \Theta| \geq \varepsilon) \geq P(\widetilde{\Theta}_2' > \Theta + \varepsilon) = P(X_{min} \cdot (n+1) \geq \Theta + \varepsilon) =$$

$$= P\left(X_{min} \geq \frac{\Theta+\varepsilon}{n+1}\right) = \left(1 - \frac{\Theta+\varepsilon}{n+1} \cdot \frac{1}{\Theta}\right)^n \xrightarrow[n\to\infty]{} e^{-\frac{\Theta+\varepsilon}{\Theta}} > 0$$

для равномер. распред. $F(x) = \frac{x}{\Theta}$, $0 \leq x \leq \Theta$

$\Rightarrow \widetilde{\Theta}_2'$ не явл. состоят.

$$P(X > a) = 1 - P(X \leq a) = 1 - \frac{a}{\Theta}$$

Для Θ_2

$$P(|\Theta_2 - \Theta| \geq \varepsilon) = P(X_{min} \leq \Theta - \varepsilon) = P(X_{min} < \Theta - \varepsilon) + \underbrace{P(X_{min} = \Theta - \varepsilon)}_{0} =$$

$$= 1 - P(X_{min} \geq \Theta - \varepsilon) = 1 - \left(1 - \frac{\Theta - \varepsilon}{\Theta}\right)^n = 1 - \left(\frac{\varepsilon}{\Theta}\right)^n \xrightarrow[n\to\infty]{} 1 \neq 0$$

$$\Uparrow$$

$$[P(X_1 \geq \Theta - \varepsilon)]^n$$

$\Rightarrow \Theta_2$ не явл. состоят.

$$P(X_1 \geq \Theta - \varepsilon) = 1 - P(X_1 < \Theta - \varepsilon) = 1 - \frac{\Theta - \varepsilon}{\Theta} = \frac{\varepsilon}{\Theta}$$

3) $\widetilde{\Theta}_3 = X_{max}$

$$\xi_{max} \sim \underbrace{(F(x))^n}_{\Psi(x)}$$

$$\varphi(x) = \Psi(x)' = n \cdot \left(\frac{x}{\Theta}\right)^{n-1} \cdot \frac{1}{\Theta} \cdot I\{(0,\Theta)\}$$

$$M X_{max} = \int_0^\Theta x \cdot n \cdot \left(\frac{x}{\Theta}\right)^{n-1} \cdot \frac{1}{\Theta}\, dx = \frac{n}{\Theta^n} \cdot \frac{\Theta^{n+1}}{n+1} = \frac{n}{n+1}\Theta \text{ — смещенная}$$

$$\widetilde{\Theta}_3' = \frac{n+1}{n} \cdot X_{max} \qquad M\widetilde{\Theta}_3' = \frac{n+1}{n} \cdot M X_{max} = \frac{n+1}{n} \cdot \frac{n}{n+1}\Theta = \Theta$$

несмещенная

$$D\tilde{\theta}_3 = D\left(\frac{n+1}{n}\cdot X_{max}\right) = \left(\frac{n+1}{n}\right)^2 \cdot D X_{max}$$

$$M X_{max}^2 = \int_0^\theta x^2 \cdot n \cdot \left(\frac{x}{\theta}\right)^{n-1} \cdot \frac{1}{\theta}\, dx = \frac{n}{\theta^n}\int_0^\theta x^{n+1}\, dx = \frac{n}{\theta^n}\left(\frac{x^{n+2}}{n+2}\right)\Big|_0^\theta =$$

$$= \frac{n}{\theta^n}\cdot\frac{\theta^{n+2}}{n+2} = \frac{n\theta^2}{n+2}$$

$$D X_{max} = M X_{max}^2 - (M X_{max})^2 = \theta^2\left(\frac{n}{n+2} - \frac{n^2}{(n+1)^2}\right) = \frac{\theta^2 n}{(n+1)^2(n+2)}$$

$$\Rightarrow D\tilde{\theta}_3 = \frac{(n+1)^2}{n^2}\cdot\frac{n\theta^2}{(n+1)^2(n+2)} = \frac{\theta^2}{n(n+2)} \xrightarrow[n\to\infty]{} 0 \Rightarrow \text{сост. оценка}$$

$$\forall\theta>0\ \ \forall\varepsilon>0\ \ P(|\tilde{\theta}_3 - \theta| \geq \varepsilon) \xrightarrow[n\to\infty]{} 0$$

$$P\left(\left|\frac{n+1}{n}\cdot X_{max} - \theta\right| \geq \varepsilon\right) = P\Big(\underbrace{\frac{n+1}{n}}_{c} X_{max} \geq \theta+\varepsilon\Big) + P\left(\frac{n+1}{n}\cdot X_{max} \leq \theta-\varepsilon\right)$$

$$X \in [0,\theta) \Rightarrow X_{max} \geq \frac{\theta+\varepsilon}{c}$$

$$\frac{\theta+\varepsilon}{c} \geq \theta \Rightarrow \theta+\varepsilon \geq \theta\cdot c \Rightarrow \varepsilon \geq \theta(c-1) = \theta\cdot\frac{1}{n}$$

$\Rightarrow$ если $n \geq \frac{\theta}{\varepsilon}$, то $\frac{\theta+\varepsilon}{c} \geq \theta$

$\Rightarrow$ для таких n $X_{max} \geq \frac{\theta+\varepsilon}{c}$ – невозможно, т.к. $X_{max} \leq \theta$

$\Rightarrow P(c\cdot X_{max} \geq \theta+\varepsilon) = 0$, для $n \geq \frac{\theta}{\varepsilon}$

$$P\left(X_{max} \leq \frac{\theta-\varepsilon}{c}\right) = \left(\frac{\theta-\varepsilon}{c\cdot\theta}\right)^n = \left(1-\frac{\varepsilon}{\theta}\right)^n\cdot\left(\frac{1}{c}\right)^n$$

$$\left(\frac{1}{c}\right)^n = \left(1-\frac{1}{n+1}\right)^n \xrightarrow[n\to\infty]{} e^{-1}$$

$$\left(1-\frac{\varepsilon}{\theta}\right)^n \xrightarrow[n\to\infty]{} 0$$

$$\Rightarrow \lim_{n\to\infty} P(|\tilde{\theta}_3 - \theta| > \varepsilon) = 0 \quad \text{– состоятельная}$$

$$\forall\theta>0\ \ \forall\varepsilon>0\ \ P(|\theta_3 - \theta| \geq \varepsilon) \xrightarrow[n\to\infty]{} 0$$

$$P(|X_{max} - \theta| \geq \varepsilon) = P(X_{max} \leq \theta-\varepsilon) + \underbrace{P(X_{max} \geq \theta+\varepsilon)}_{0}$$

т.к. $X \leq \theta$

$$P(X_{max} \leq \theta-\varepsilon) = \left(\frac{\theta-\varepsilon}{\theta}\right)^n = \left(1-\frac{\varepsilon}{\theta}\right)^n \xrightarrow[n\to\infty]{} 0$$

$$\Rightarrow \forall\theta>0\ \ \forall\varepsilon>0\ \ P(|X_{max} - \theta| \geq \varepsilon) \xrightarrow[n\to\infty]{} 0 \quad \text{– состоятельная}$$

4) $\tilde{\theta}_4 = X_1 + \frac{1}{n-1}\sum_{i=2}^{n} X_i$

$$M\tilde{\theta}_4 = M X_1 + \frac{1}{n-1}\sum_{i=2}^{n} M X_i = M\xi + \frac{1}{n-1}\cdot(n-1)M\xi = \frac{\theta}{2} + \frac{\theta}{2} = \theta$$

несмещенная

$$D\tilde{\theta}_4 = D X_1 + D\left(\frac{1}{n-1}\sum_{i=2}^{n} X_i\right) = D\xi + \left(\frac{1}{n-1}\right)^2\sum_{i=2}^{n} D X_i = D\xi + \frac{1}{(n-1)^2}\cdot(n-1)\cdot D\xi =$$

$$= \frac{\theta^2}{12} + \frac{1}{n-1}\cdot\frac{\theta^2}{12} \xrightarrow[n\to\infty]{} 0$$ (достат. усл. не выполнено) состоятельность

По ЗБЧ Хинчина: ξ_i - независ., одинаковое распределение, $\exists M\xi_i < \infty$

$$\Rightarrow \underbrace{\frac{1}{(n-1)}\sum_{i=2}^{n} \xi_i}_{=\tilde{\theta}_4'} \xrightarrow{P} M\xi = \frac{\theta}{2}$$

$$MX_1 = M\xi_1 = \frac{\theta}{2}$$

Пусть $\tilde{\theta}_4 \xrightarrow{P} \theta$ состоятельна

$$\Rightarrow \tilde{\theta}_4 - \tilde{\theta}_4' \xrightarrow{P} \theta - \frac{\theta}{2} = \frac{\theta}{2}$$

Но $\tilde{\theta}_4 - \tilde{\theta}_4' = X_1 \Rightarrow X_1 \xrightarrow{P} \frac{\theta}{2}$

$$\Downarrow$$

$P(|X_1 - \frac{\theta}{2}| \geq \varepsilon) \xrightarrow[n\to\infty]{} 0$, но вер-ть не зависит от n

$$\Rightarrow P(|X_1 - \frac{\theta}{2}| \geq \varepsilon) = 1 - \frac{2\varepsilon}{\theta}$$

$\Rightarrow \tilde{\theta}_4$ не состоятельна